MQGT–SCF Hardening Completion Update (2026)

Specification-to-Referee Upgrade Summary

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Abstract

This document records the scientific hardening pass applied to the 2026 MQGT–SCF Theory of Everything stack. The update does not claim new empirical confirmation; it upgrades formal and publication readiness by converting prior narrative closure into theorem-oriented, operational, and audit-ready structure. The completed upgrades are: (i) canonical spine consolidation, (ii) no-signaling theorem package v2, (iii) operational ethics estimator definitions for $e(x,t)$, $E_R(t)$, and ΔE_a, (iv) parameter hygiene with identifiability/bound tags, (v) one flagship falsifier with explicit sensitivity threshold, (vi) canonical matter/brain source-term model, and (vii) a reviewer-proof release checklist. Remaining uncertainty is empirical: parameter fitting, hostile replication, and independent experimental validation.

1 Purpose

The practical problem was not conceptual incompleteness of the framework, but publishability drift: multiple versions of definitions, mixed theorem/prose standards, and non-uniform release gates. This update enforces one canonical scientific path from formal equations to falsification protocol.

2 Completed hardening upgrades

2.1 Canonical spine consolidation

The narrative source of truth was consolidated into a single solved-vs-hardening map with explicit A–H closure routing. This removes ambiguity about what is already closed at specification level versus what remains empirical.

2.2 No-signaling theorem package v2

The no-signaling section was upgraded from first-order narrative to theorem structure:

- assumptions block (local support, microcausality, linear CPTP, spacelike protocol separation),
- lemma chain (commuting local increments, reduced-map closure),
- theorem statement (operational no-signaling under local covariant GKSL),
- scope-boundary remark (explicit failure conditions).

This shifts the section from suggestive argumentation to auditable theorem scaffolding.

2.3 Operational ethics estimator

To eliminate interpretive freedom, the ethical weighting pathway is now explicitly computable:

$$e(x,t) = \sum_{k=1}^{K} w_k O_k(x,t), \tag{1}$$

$$E_R(t) = \int_V e(x,t)\, d^3x, \tag{2}$$

$$\Delta E_a = E_R^{(a)}(t_*) - E_R^{\text{baseline}}(t_*), \tag{3}$$

$$E_a = \Delta E_a / C_E. \tag{4}$$

The estimator is canonical for headline claims; alternatives are relegated to sensitivity analysis.

2.4 Parameter hygiene and bounds consolidation

Core and empirical manuscripts now share one parameter inventory, with each parameter tagged by: role, prior class, identifiability, and current bound source channel. This addresses the common "too many knobs" criticism by exposing parameter governance explicitly.

2.5 Flagship falsifier

One primary falsifier is now fixed as interferometric visibility suppression:

$$\frac{V}{V_0} \approx \exp(-\Gamma T \Delta x^2), \quad \Gamma \lesssim \frac{|\ln(1-\delta)|}{T\Delta x^2} \tag{5}$$

for null precision target δ. This creates a single quantitative exclusion blade with direct sensitivity mapping.

2.6 Canonical matter/brain coupling

Main-text coupling drift was removed by fixing a canonical source model:

$$\mathcal{O}_m(x) = w_n n(x) + w_s s(x), \quad w_n, w_s \geq 0,\ w_n + w_s = 1 \tag{6}$$

with canonical source terms

$$J_{\Phi_c} = g_{\Phi_c}\mathcal{O}_m + \alpha_H |H|^2, \quad J_E = g_E \mathcal{O}_m + \beta_H |H|^2. \tag{7}$$

Alternative couplings are appendix-only.

2.7 Reviewer-proof release gate

A dedicated reviewer-proof checklist now enforces theorem completeness, notation consistency, falsifier clarity, reproducibility pinning, and release hygiene before publication claims.

3 Scientific effect of the update

This hardening pass improves the framework in three concrete ways:

1. **Mathematical auditability:** proof-carrying structure replaces prose-only arguments.

2. **Operational testability:** ethics and falsifier definitions are now executable and preregisterable.

3. **Publication robustness:** canonicalization and checklist gating reduce reviewer-facing drift.

Importantly, these are quality-of-theory upgrades, not new empirical confirmations.

4 What remains external

The remaining high-stakes uncertainty is empirical and cannot be closed by prose:

- independent replication and hostile reanalysis,
- stronger no-signaling stress tests in edge entanglement settings,
- tighter parameter posteriors from new data,
- flagship falsifier execution at declared sensitivity.

5 Canonical artifact map

Artifact	Canonical location
Narrative spine	`docs/CANONICAL_SPINE.md`
Index and routing	`docs/INTEGRATED_MANUSCRIPT_INDEX.md`
Core theorem spine	`papers_sources/MQGT_SCF_Minimal_Consistent_Core_2026.tex`
Empirical bounds layer	`papers_sources/MQGT-SCF_Minimal_Consistent_Core_Empirical_Vali`
Teleology companion	`papers_sources/Teleology_Covariant_Boundary_Selection_...2026.`
Falsification protocol	`docs/FALSIFICATION_PACKET.md`
Release gate	`docs/REVIEWER_PROOF_CHECKLIST.md`

6 Conclusion

The framework is now materially stronger in scientific form: one canonical spine, theorem-oriented no-signaling, operational ethics estimator, explicit parameter governance, one primary falsifier, canonical coupling model, and reviewer-proof release gates. This is the correct bridge from specification-level closure to referee-grade physics readiness, with empirical closure still delegated to experiment.